

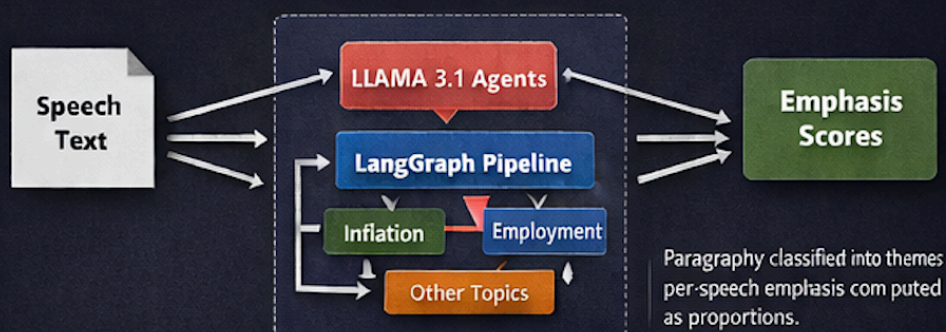
Agentic NLP for Thematic Analysis of Fed Speeches: Yield Impacts under Jerome Powell (2018–2025)

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Motivation

- Do thematic focuses in Jerome Powell's Fed speeches affect U.S. Treasury yields?
- Powell era (2018–2025) saw heightened market sensitivity to Fed communication.
- Need for granular NLP analysis of central bank speeches.

Agentic NLP Framework



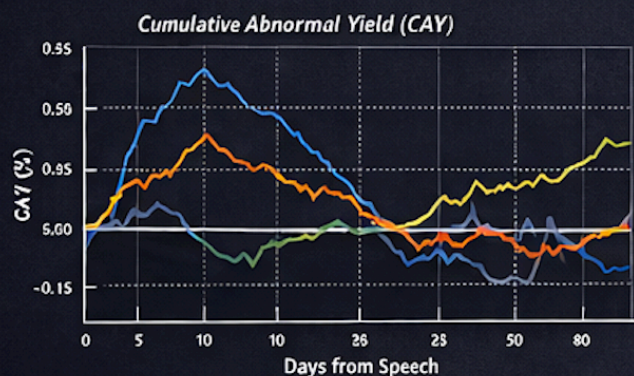
Data & Methodology

- 1,200+ speeches scraped from the Fed's official archive
- LangGraph pipeline orchestrating LLAMA 3.1 agents for paragraph tagging.
- Themes, Inflation / Employment / Other $\rightarrow$ per-speech emphasis proportions.
- Event study: estimation window [-120, -10], event window [0, +].
- Regression: $CAY = \text{inflation_emphasis} + \text{employment_emphasis} + \text{FedFundsRate}$.

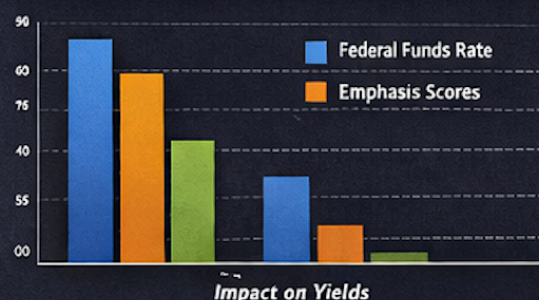
Key Findings

- Federal Funds Rate is the dominant predictor of yield changes.
- Inflation & Employment emphasis scores show no statistically significant direct yield impact.
- Narrative framing done appears insufficient to move long horizon yields.

Event Study Analysis



Influence Factors



Conclusion & Future Work

- Agentic NLP provides a scalable, reproducible way to quantify paragraph-level themes in central-bank communication.
- Null result is informative: bond markets may price policy stance more than narrative emphasis in speeches.
- Next: sentiment-tone agents, hierarchical paragraph weighting, compositional transforms (ILR), and robustness with HAC/bootstopper SEs.

Code & Data



Reproducible pipeline, data, and diagrams

<https://github.com/pankajti/ecai2025>

<https://www.linkedin.com/in/pankajti/>